MEDLOG-PH: An LGU-Based Web System for Medicine and Medical Relief Logistics Coordination

Christian Ivan A. Los Baños

Bachelor of Science in Information Technology

St. Dominic College of Asia

Abstract

This paper presents MEDLOG-PH, a web-based medicine and medical relief delivery coordinator designed for a Philippine local government unit (LGU)-centered health and disaster response setting. The system addresses common coordination problems in medicine requests, inventory monitoring, donation verification, logistics tracking, proof documentation, analytics, and report generation. A developmental research design and Agile iterative prototyping were used to develop the system using Next.js, Firebase Authentication, Cloud Firestore, UploadThing, Chart.js, and Vercel. Synthetic data was used to simulate realistic Philippine LGU, hospital or health center, donor, inventory, and logistics transactions without using confidential patient, donor, or government records. The system demonstrated the capacity to centralize fragmented workflows and support transparent health-relief coordination. Actual pilot testing with LGU personnel, health facilities, donors, logistics staff, and IT experts is recommended before full deployment.

Keywords: MEDLOG-PH, LGU logistics, medical relief, synthetic data, disaster response, Philippine health information system

Introduction

The Philippines is frequently affected by typhoons, flooding, earthquakes, volcanic hazards, and public health emergencies, making timely coordination of medicines and relief goods important for LGUs and health facilities. Philippine disaster policy gives LGUs an essential role in preparedness, response, coordination, and information sharing, while recent disaster response reports show that

emergency medicines and logistics are regularly deployed to affected regions. In this context, medicine and medical relief logistics require a system that can connect requests, inventory, donations, delivery updates, and supporting documents in one workflow.

Many local offices still depend on manual forms, spreadsheets, phone calls, messaging applications, and separate records. These practices may cause delayed approval, inaccurate stock counts, weak donation traceability, and limited visibility of delivery status. MEDLOG-PH was therefore developed as a centralized web-based platform for a Philippine LGU-centered health and relief coordination office. It provides role-based access for administrators, hospital or health center staff, donors, and logistics personnel, and it supports request submission, approval workflow, inventory monitoring, donation verification, delivery tracking, proof upload, analytics, and report generation.

Methodology

The study used a developmental research design because the main output was a working web-based information system. Agile iterative prototyping guided planning, interface design, module development, integration, testing, revision, and deployment preparation. The development process identified the coordination needs of an LGU health and relief office, then translated these needs into modules for authentication, requests, inventory, donations, deliveries, analytics, and reports. The technology stack included Next.js for the frontend, Firebase Authentication and Cloud Firestore for identity and database services, UploadThing for file uploads, Chart.js for dashboards, and Vercel for deployment.

The study used synthetic data rather than actual LGU, hospital, donor, or patient records. Synthetic data refers to artificial but realistic sample records created to simulate operational scenarios. In MEDLOG-PH, this included sample medicine requests, inventory quantities, donation records, delivery assignments, proof uploads, user roles, and evaluation responses. The purpose was to test system functions, demonstrate realistic workflows, generate reports and dashboard

outputs, and avoid privacy risks while the system is still in prototype validation. This approach is consistent with ethical handling of sensitive information because real deployment may involve personal information, organizational records, health-related transactions, and uploaded proof documents.

Results and Discussion

The completed system demonstrated the major functions needed for Philippine LGU-centered medical relief coordination. Hospital or health center users can submit medicine or relief requests and monitor status. Administrators can approve or reject requests, check inventory levels, verify donations, assign logistics personnel, view analytics, and generate reports. Donors can submit donation details and upload proof. Logistics personnel can update deliveries from preparing to in transit to delivered and upload proof of delivery. These functions match the need for supply visibility and coordinated health logistics reflected in Philippine eLMIS and supply chain initiatives.

Synthetic testing showed that MEDLOG-PH can reduce fragmented documentation by linking requests, inventory, donations, and deliveries into one digital workflow. Approved requests deduct items from inventory, verified goods donations update stock, and delivery records remain connected to the approved request. The analytics dashboard supports decision-making by showing request trends, donation statistics, delivery completion, and inventory status. This is useful for LGU settings where decisions must be made quickly during floods, typhoons, disease outbreaks, evacuation center operations, or community health emergencies.

Using selected software quality criteria, the synthetic evaluation indicated very high acceptability in functionality, usability, reliability, security, maintainability, and portability. Functionality was supported by the completed request-to-report workflow. Usability was supported by role-based dashboards and clear status labels. Reliability was observed through repeated synthetic transactions. Security was supported by authentication and role restrictions, while maintainability and portability were supported by modular development and browser-based access. However, the

results should be treated only as system-testing evidence because synthetic data does not replace actual user validation.

Conclusion and Recommendations

MEDLOG-PH demonstrates how a web-based logistics coordinator can support Philippine LGU health and disaster response operations by centralizing medicine requests, inventory monitoring, donation verification, delivery tracking, proof uploads, analytics, and reports. Synthetic data allowed safe and realistic testing without exposing confidential records. The system is suitable for capstone-level prototype evaluation because it applies software development to a relevant public-service problem in the Philippine context. Actual pilot validation is still necessary to confirm usability, workflow fit, data privacy readiness, and operational effectiveness in a real LGU or health facility setting.

It is recommended that the next phase include pilot testing with LGU administrators, health center or hospital representatives, donors, logistics personnel, and IT experts. The pilot test should include informed consent, data privacy notices, official permission from the target office, and technical review for security, backup and restore, browser compatibility, and performance. Future versions may add SMS or email alerts, QR-coded inventory release, medicine expiration alerts, GPS-based delivery monitoring, audit trail export, and integration with official LGU report formats.

References

Center for Excellence in Disaster Management and Humanitarian Assistance. (2025). Philippines disaster management reference handbook. PreventionWeb. https://www.preventionweb.net/publication/documents-and-publications/philippines-disaster-management-reference-handbook-2025

Department of Health. (2024). Department Memorandum No. 2024-0309: National Strategic Plan for Integrated Supply Chain Management 2024-2028. Health Facility Development Bureau. https://sites.google.com/view/doh-hfdb/2024-q3-updates/dm-2024-0309

Elepaño, A., Sy, T. R. L., Ulanday, M. E. D., Chan, V. F., Reyes, K. A. V., Isip-Tan, I. T., Estuar, M. R. J. E., & Dans, A. L. (2025). Implementing electronic health records in Philippine primary care settings: Mixed-methods pilot study. JMIR Medical Informatics, 13, e63036. https://doi.org/10.2196/63036

Evio, B. D., & Bonito, S. R. (2024). Formative evaluation of the implementation of eHealth in the Philippines: A qualitative study. Acta Medica Philippina, 58(12), 35-47. https://doi.org/10.47895/amp.v58i12.9289

Hermosura, M. J. E., Decendario, A. C. C., & Chiong, L. A. (2025). The Philippine disaster risk reduction and management in health: A sequential explanatory approach. Journal of Interdisciplinary Perspectives, 3(8). https://doi.org/10.69569/jip.2025.404

Medicines, Technologies, and Pharmaceutical Services Program. (2024). Strengthening health commodity procurement and supply chain management in the Philippines. USAID MTaPS. https://www.mtapsprogram.org/our-resources/strengthening-health-commodity-procurement-and-supply-chain-management-in-the-philippines/

National Disaster Risk Reduction and Management Council. (2024). Updated National Disaster Risk Reduction and Management Plan 2020-2030. ReliefWeb. https://reliefweb.int/report/philippines/philippines-national-disaster-risk-reduction-and-management-plan-2020-2030

Philippine News Agency. (2025). DOH sends P31-M worth of emergency medicines, logistics. https://www.pna.gov.ph/articles/1254838

Republic Act No. 10121, Philippine Disaster Risk Reduction and Management Act of 2010. (2010). Supreme Court E-Library. https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/21121

Republic Act No. 10173, Data Privacy Act of 2012. (2012). National Privacy Commission. https://privacy.gov.ph/data-privacy-act/

Site and Users

Deployed Site: https://medlog-ph.vercel.app

Admin Username: christian@gmail.com

Admin Password: christian123

Hospital Staff Username: dennis@gmail.com

Hospital Staff Password: dennis123

Logistics Username: jastin@gmail.com

Logistics Password: jastin123

Donor Username: mareya@gmail.com

Donor Password: mareya123